Sathyakumar Nanda

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Executive Summary

Deep Learning engineer and researcher with a completed Master's in Robotics & AI (Warsaw University of Technology) and an ongoing Master's in Computational Cognitive Science & ML (University of Warsaw). Research spans multi-agent reinforcement learning, RLHF, and agentic LLM systems, with both academic and commercial deployment experience. Driven by the intersection of how systems learn and how intelligence works in practice. Currently completing final semester; available for research roles and internships.

https://github.com/sathyakumarnandakumar-cmyk

Skills

Deep Learning & ML: PyTorch · TorchRL · LangChain · LangGraph · Gymnasium · RAG / Semantic Retrieval · Reinforcement Learning (MARL · HRL · RLHF) · Diffusion Models · Vision Transformers (ViT) · Hyperparameter Tuning (Optuna) · Experiment Tracking (W&B) · Distributed Training (DDP · FSDP) · HPC & Cluster Computing (SLURM) **Infrastructure & DevOps:** FastAPI · Redis · Docker · Terraform · GCP · Linux · Bash | **Programming:** Python · C++ · SQL · Git

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Education

Warsaw University of Technology

Masters in Robotics and Automatic Control • Warsaw 02/2022

GPA: 4.5/5. - Graduated. Secured a NAWA- Banacha scholarship for one semester.

MIM + Psych, University of Warsaw*

Masters in Computational Cognitive Science and ML (Current) • Warsaw

Specilisation: Machine Learning

- *Includes an **Erasmus+ exchange for two semesters to Uni. of Copenhagen**, Denmark.*

Related Professional Experience

Masters Thesis- Co-Existence Project - Prof. Rafal Kucharski

Academic Researcher - Masters Thesis | MARL . • GMUM, Uni Jagellonian

Urban Routing Benchmark Extension: Hierarchical RL for Autonomous Urban Routing using Multiple Agent RL

- Earlier research URB benchmark (NeurIPS 2025) indicates that no SOTA conventional MARL algorithm like iPPO, QMix consistently outperformed human-baseline heuristics, exposing a critical scalability gap in cooperative routing
- Extracted dynamic traffic features directly from the SUMO/TraCI environment (speeds, densities, queue lengths, occupancy) and engineered enriched observation vectors including agent position encodings, increasing contextual awareness for MARL agents
- Designed a Hierarchical RL architecture with a meta-controller issuing subgoals to low-level routing agents, targeting the scalability gap exposed by the original benchmark
- Demonstrated promising convergence; ongoing work targets robustness across diverse network topologies

Freelance NLP/ Agentic Engineer for a commercial Project

Agentic NLP System for Real-Time Financial Intelligence | Python · LangGraph · LangChain · FastAPI · Redis

- Built and commercially licensed a product to a leading Indian financial data provider - Architected a LangGraph-based multi-agent system of ~10 specialized agents with ~50 modular tool-calling skills; implemented reducer-driven state schemas, persistent checkpointing, conditional edge routing, and scatter-gather parallel execution patterns for robust multi-agent orchestration at production scale
- Engineered chain-of-thought reasoning pipelines with structured prompt engineering for financial feature extraction and causality analysis — agents perform grounded reasoning over live data via semantic retrieval and agentic web search
- Built a RAG-style event analysis pipeline where agents retrieve, reason over, and synthesize real-world market narratives from live web sources to explain anomalous signals in structured numerical data
- Engineered chain-of-thought reasoning pipelines with structured prompt engineering for financial feature extraction and causality analysis — agents perform grounded reasoning over live data via semantic retrieval and agentic web search
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University of Warsaw

Student Researcher

Notable academic projects:

- **Quantization-Aware FSDP for large scale Fine-tuning**- Implement a training pipeline that integrates 4-bit/8-bit quantization with FSDP's mixed_precision policy. Analyze the gradient synchronization overhead and accuracy trade-offs when using FSDP to shard quantized weights for memory-constrained fine-tuning of a Llama-3 (70B) scale model.

TensorCell Research Group- Dr. Pawel Gora

Independent Researcher

Multi-Agent RL for Urban Traffic Signal Control

- Implemented and tested a Multi-agent Reinforcement learning framework for urban traffic signal control.
- We found that the standard SOTA networks like A2C/ iPPO fail to achieve optimal state for complex traffic networks easily and is very sensitive to parameters. We are developing and testing specific reward models and more centralised algorithms like Mixed Actor-critic networks to improve generalisation across networks.
- Introduced a Dyna-inspired experience replay memory buffer to address prohibitively slow convergence in traditional Urban RL training pipelines, significantly accelerating sample efficiency and convergence speed.

Warsaw Global Robotics Hackathon

Team Member • Lute, Warsaw

Imitiation Learning and Vision Language Action Models for Robot manipulation

- Fine-tuned π0.5 via LeRobot to map RGB-D streams + language prompts to continuous joint-space actions for a three day hackathon based on VLAs in robots.
- Implemented diffusion head tuning and data augmentation optimization within the 24-hr online training time window, achieving a **20% higher task success rate** than the competition baseline.

EiT, Warsaw University of Technology

Student Academic Researcher • Warsaw

Human-Object Interaction (HOI) for the TIAGo Service Robot in Domestic Settings

- Developed a robust object detection codebase trained on a proprietary dataset of common household items and common human action commands to social robots.
- Engineered a real-time human action recognition system by fusing lightweight skeletal pose data with embeddings producted by other action recognition models running inference in parallel. With this method, the training/ optimisation time was greatly reduced for custom datasets.
- Streamlined the inference pipeline to ensure low-latency performance, enabling responsive social interactions in dynamic environments.

Research Journal & Conference Publications

1. **A Build-Your-Own Three Axis CNC PCB Milling Machine;** *Materials Today: Proceedings, Volume 5 · Dec 7, 2018*
2. **A build your own robot manipulator for manufacturing education;** *6th International Conference on Computer Applications in Electrical Engineering-Recent Advances (CERA) · Nov 4, 2017*

Related Courses and Certificates

University of Warsaw + Erasmus @University of Copenhagen: Natural Language Processing, Statistics, Data Science, Large Scale Machine Learning, Building RAG agents with LLMs, Reinforcement Learning. **Certifications:** Deep Learning using Keras and Tensorflow, Intermediate Deep Learning with PyTorch. Real time OS with Linux. Nvidia- Building RAGs for LLMs, Building AI agents using Langgraph.

Special Mentions

- **Co-Editor, Weekly Newsletter @ Warsaw.ai** — Translate complex AI publications and technical breakthroughs into accessible, high-impact summaries for a diverse professional audience.
- **Warsaw Global Robotics Hackathon @ Lute Corp** — Winning team, LeRobot YARM Challenge.
- **Warsaw.ai Hackathon** — Finalist, Assesco RAG Systems Challenge.